

ROSECRANS, MO, USA

WMO: 724490

Lat: 39.774N

Lon: 94.923W

Elev: 818

StdP: 14.27

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 13993

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.6	5.7	-7.7	3.7	1.6	-2.8	4.9	7.6	28.4	28.2	26.0	29.8	7.0	320	0.532

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	94.5	77.1	92.0	76.4	90.0	75.6	80.1	90.8	78.4	88.9	77.1	87.1	11.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.0	145.0	87.1	75.2	136.2	85.4	73.6	129.0	83.6	44.2	90.5	42.6	89.6	41.0	86.7	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.7	22.7	19.7	-7.0	98.5	5.8	2.9	-11.1	100.6	-14.5	102.3	-17.8	104.0	-22.0	106.1		
			-7.6	82.7	5.5	1.0	-11.5	83.4	-14.7	84.1	-17.8	84.6	-21.8	85.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.2	28.0	31.9	43.5	54.5	65.2	74.4	77.8	75.5	68.3	55.7	42.8	31.1
	DBStd	19.91	11.67	11.87	11.44	9.89	8.41	6.13	5.56	5.77	7.72	9.59	10.11	11.09
	HDD50	2430	684	514	266	64	3	0	0	0	1	50	257	591
	HDD65	5280	1148	926	671	337	104	6	1	2	53	315	666	1051
	CDD50	3958	2	8	64	199	475	732	862	792	550	227	42	5
	CDD65	1332	0	0	4	21	111	288	398	329	153	27	1	0
	CDH74	13653	0	0	41	242	1082	2922	4287	3253	1529	288	9	0
	CDH80	5468	0	0	4	58	351	1161	1907	1371	542	74	0	0
Wind	WSAvg	8.6	9.1	9.4	10.5	10.9	9.4	8.5	6.6	6.2	7.0	8.1	9.0	8.8
Precipitation	PrecAvg	35.2	0.9	1.0	2.3	3.0	4.8	4.7	3.7	4.3	4.5	2.9	1.7	1.4
	PrecMax	55.6	3.5	3.9	7.3	6.7	11.5	11.9	9.6	10.4	12.2	8.2	4.3	4.8
	PrecMin	20.7	0.0	0.0	0.3	0.9	1.3	1.0	0.0	0.4	0.8	0.0	0.0	0.0
	PrecStd	9.0	0.8	0.8	1.7	1.7	2.4	2.6	2.8	2.7	3.5	2.1	1.4	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.3	70.1	79.2	86.5	92.6	95.7	98.8	98.7	93.3	87.4	75.0	64.2
		MCWB	49.3	54.2	61.9	67.2	72.0	77.1	77.9	76.5	74.0	69.7	60.0	57.3
	2%	DB	55.9	62.2	73.0	80.7	87.8	92.2	94.8	93.3	89.9	81.4	69.8	57.3
		MCWB	46.7	51.7	58.5	63.5	69.5	75.6	78.3	77.3	74.1	65.3	58.6	50.3
	5%	DB	50.0	55.3	67.9	75.6	83.7	89.8	92.1	90.6	86.2	76.6	64.0	52.3
		MCWB	43.0	46.5	56.5	61.2	68.1	74.9	77.8	76.4	72.5	63.6	53.9	46.0
	10%	DB	44.5	49.9	63.1	72.0	80.5	86.9	89.8	87.9	82.3	72.4	59.6	47.0
		MCWB	38.9	43.0	53.5	59.7	67.0	73.2	76.9	75.3	69.5	62.1	51.4	41.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.3	58.6	65.4	69.3	75.3	80.2	82.5	81.2	77.6	72.8	63.1	59.0
		MCDB	59.4	64.3	75.2	80.8	88.0	90.6	92.3	91.4	89.3	81.6	70.8	63.5
	2%	WB	47.1	52.5	62.1	66.3	72.9	78.0	80.7	79.3	75.6	69.5	59.3	52.3
		MCDB	55.2	59.0	69.4	76.2	84.0	89.4	91.1	90.3	86.1	77.0	66.7	55.5
	5%	WB	43.2	47.4	58.7	63.7	70.8	76.3	79.3	77.7	74.0	66.4	56.3	46.3
		MCDB	50.2	56.8	66.5	73.5	80.8	87.1	89.9	88.1	83.5	73.9	62.9	51.1
	10%	WB	38.8	42.9	53.4	60.5	68.6	74.7	77.8	76.2	72.2	63.0	52.4	42.1
		MCDB	44.1	49.5	62.8	70.0	77.5	84.4	87.3	85.7	80.3	70.2	58.7	47.1
Mean Daily Temperature Range	5% DB	MDBR	19.3	19.6	21.6	22.3	20.9	19.7	19.4	20.6	23.5	23.4	21.3	18.6
		MCDBR	28.6	29.5	29.6	28.2	25.3	21.8	21.7	23.8	25.0	28.2	28.8	26.2
		MCWBR	20.2	19.6	17.3	15.4	11.3	9.6	9.3	10.9	11.4	14.8	18.0	19.4
	5% WB	MCDBR	27.0	27.1	24.1	24.1	21.1	19.6	19.2	20.7	21.6	22.2	24.7	23.8
		MCWBR	19.7	19.2	15.3	15.0	11.1	9.9	9.7	10.6	10.8	13.6	17.4	18.9
Clear-Sky Solar Irradiance	taub	0.292	0.306	0.336	0.386	0.419	0.432	0.434	0.433	0.386	0.340	0.302	0.288	
	taud	2.454	2.424	2.404	2.284	2.255	2.254	2.248	2.271	2.353	2.469	2.516	2.487	
	Ebn at Noon	279	291	292	282	273	269	267	264	271	274	274	272	
	Edh at Noon	25	30	34	41	43	43	43	41	36	28	23	23	
All-Sky Solar Radiation	RadAvg	677	946	1226	1560	1819	2040	2036	1798	1508	1062	746	583	
	RadStd	62	83	107	99	156	153	139	87	102	119	65	58	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page